Optimizing Inventory Management and Strategizing Expansion of Corindo Spices

Midterm Report for the BDM Capstone Project

KHUSHI PANDEY

23f1002421

IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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1. Executive Summary

Corindo Spices is a regional wholesale spice business located in Rajbandigarh, Gouripur, West Bengal, aiming to grow its business across the state.

The business is facing two major challenges. The first is an inventory management issue, particularly while launching a new product. The second challenge is its limited experience in entering new markets, where it must deal with the competition from established brands.

This report presents a comprehensive analysis of Corindo Spices, focusing on sales patterns, pricing strategies, and inventory management to support the company's plans. Key findings indicate that spice sales are generally influenced by seasonal trends, most spices see higher demand during festive seasons.

The analysis also explores initial stock calculations, forecasting demand using regression models, and ideal price points for upcoming products. Inventory planning incorporates safety stock percentages to prevent stockouts, while moving average analyses provide insights into short-term demand fluctuations.

Overall, the report highlights actionable insights for Corindo Spices to optimize pricing, align inventory with demand, and strategically plan product launches. By addressing seasonal trends, anomalous sales patterns, and inventory management challenges, Corindo Spices can strengthen its business and successfully expand its presence.

2. Proof of Originality of the Data

- Video Proof: Interview conducted with the business owner of Corindo Spices. Google drive link to the video can be found here
- Data Proof: Link to the excel file provided by the business can be viewed here
- Images: A few images were clicked during the visit to the factory of the business- link
- NOC: The NOC letter signed by the business-link
- Email from the business: screenshot
- Copy of dataset used for analysis: link
- Colab notebook used for analysis: link

3. Metadata

The analysis of Corindo Spices was conducted using a structured Excel dataset spread across 12 sheets, each representing one month of data from April 2024 to March 2025. Each sheet captured detailed comparisons between order quantity and sales quantity across 8 spice products, 10 districts, and 3 product sizes.

Column Headers (consistently used across sheets):

- DISTRICT
- POPPY SEEDS, CUMIN SEEDS, etc. (Products)
- 5 or 10mrp pck, 50gm pck, 500gm pck (Sizes)
- Orders
- Sales

Attribute	Description
Number of Sheets	12 (one for each month)
Sheet Names	APRIL 24, MAY 24, …, MARCH 25
Districts	10 districts per sheet
Products	8 products per sheet
Pack Sizes	3 sizes: 5MRP, 50g, 500g
Columns	District, Product, Size, Orders, Sales
No. of Rows	13 rows per sheet
Data Types	District, Product, Size → Text Orders, Sales → Integer
Notes	Each sheet represents one month, each row represents a unique combination of product, size, and district

4. Descriptive Statistics:

A brief description of the terms used is as follows:

- **Sum**: The total order/sales quantity for each product.
- **Mean**: The average order/sales quantity for a given product.
- **Standard Deviation (SD)**: The extent of variation from the mean, indicating consistency or volatility.
- **Minimum (Min) & Maximum (Max)**: The lowest and highest observed order/sales quantities.

For mini packs (MRP 5 or 10):

Product	Orders					Sales				
	Sum	Mean	SD	Min	Max	Sum	Mean	SD	Min	Max
Poppy	95355	794.63	1004.20	90	3600	94045	783.708	994.419	90	3500
Cumin	43554	362.95	361.81	65	1330	42849	357.08	356.60	65	1330
Brown Mustard	23695	197.46	272.38	30	1050	23011	191.76	267.33	20	1050
Yellow Mustard	30921	257.66	359.67	30	1200	30615	255.13	356.96	30	1189
Fennel	17279	143.99	193.74	35	765	17052	142.10	192.62	32	750
Panch Phoran	52004	433.37	649.99	30	2400	51331	427.76	642.20	30	2380
Turmeric	9811	81.76	76.69	0	350	9755	81.29	75.94	0	350
Chilli	28980	241.50	356.40	0	1200	28920	241.00	356.18	0	1200

For 50g packs:

Product	Orders					Sales				
	Sum	Mean	SD	Min	Max	Sum	Mean	SD	Min	Max
Poppy	2308	19.23	9.08	8	40	2773	18.94	8.66	8	35
Cumin	6241	52.01	19.28	30	108	6109	50.91	19.08	25	100
Brown Mustard	5403	45.03	28.76	18	130	5183	43.20	28.02	10	130
Yellow Mustard	4244	35.37	24.69	0	100	4172	34.77	24.18	0	90
Fennel	5021	41.84	21.95	10	100	4899	40.83	21.28	10	95
Panch Phoran	16353	136.28	141.43	25	550	16180	134.83	140.65	25	550
Turmeric	55005	458.38	795.50	0	3500	54585	454.88	789.57	0	3200
Chilli	4850	40.42	18.57	0	100	4820	40.17	18.57	0	100

For 500g packs:

Product	Orders					Sales				
	Sum	Mean	SD	Min	Max	Sum	Mean	SD	Min	Max
Poppy	8371	69.76	52.28	15	210	8286	69.05	51.88	15	210
Cumin	8427	70.23	116.50	0	380	8371	69.76	115.71	0	380

Brown Mustard	2220	18.5	16.13	0	50	2134	17.78	15.46	0	45
Yellow Mustard	2185	18.21	15.76	0	40	2166	18.05	15.67	0	40
Fennel	2402	20.02	51.70	0	190	2360.5	19.67	50.68	0	190
Panch Phoran	29104	242.53	641.83	0	2500	28940	241.17	637.89	0	2400
Turmeric	21040	175.33	300.58	0	1150	20928	174.40	298.73	0	1100
Chilli	5225	43.54	93.32	0	380	5205	43.36	92.88	0	365

5. Explanation of Analysis Methods:

5.1. Data Collection

The dataset was collected from **Corindo Spices**, containing information on:

- **Time period:** Monthly sales data (12 months).
- **Geography:** 10 districts across the state.
- **Products:** 8 spices in 3 pack sizes.
- **Metrics:** Orders placed, and sales fulfilled.

5.2. Data Cleaning and Pre-Processing

- Some pack sizes were mislabelled (5mrp mistakenly written as 10mrp). This was corrected after verifying with the owner.
- In a few cases, sales exceeded orders. These entries were corrected in consultation with the owner.
- To enable systematic analysis, the dataset was converted into long format.

5.3. Quantitative Analysis Methods

5.3.1 ABC Classification

Tool used: Excel.

Method: Products were classified into A, B, and C categories based on their contribution to total sales.

Purpose: Guides inventory prioritization, ensures stock optimization and reduces risk of shortages in expansion areas.

5.3.2 Fulfilment Rate Analysis

Tool used: Excel.

Method: Fulfilment rate = (Sales ÷ Orders) × 100

Purpose: Indicates service quality, identifies high-performing districts. Critical for expansion decisions, areas with higher fulfilment can better support introductions.

5.3.3 District-wise Sales Analysis

Tool used: Excel.

Method: Aggregated sales across districts.

Purpose: Supports market expansion by identifying high-potential districts.

5.3.4 Moving Average Analysis

Tool used: Python (Pandas, Matplotlib).

Method: 3-month moving averages of Orders and Sales were computed.

Purpose: Helps in inventory management by forecasting expected demand and provides baseline demand insights to plan for new product launches.

5.3.5 Analogous Product Analysis

Tool used: Python, Excel.

Method: Identified existing products with similar characteristics (taste profile, price point, target market, pack sizes) to serve as analogues to estimate expected demand.

Purpose: Determine ideal price points and stock for Corindo's upcoming products (*Shahi Garam Masala*, *Kashmiri Lal Mirch*, *Coriander Powder*) based on existing competitor data.

6. Results and Findings:

ABC Analysis:

Cumulative % <70%=A, 70%-90%=B, 90%-100%=C

Product	Order Quantity	% of Total	Cumulative %	Category
Poppy	106034	22.0905	22.0905087	A
Panch Phoran	97461	20.3045	42.3949683	A
Turmeric	85856	17.8867	60.2817095	A
Cumin	58222	12.1296	72.4113434	B
Chilli	39055	8.13649	80.5478356	B
Yellow Mustarc	37350	7.78128	88.329118	B
Brown Mustarc	31318	6.52461	94.8537286	C
Fennel	24702	5.14627	100	C
Total	479998			

- The cumulative percentage distribution shows that Poppy Seeds, Panch Phoron, and Turmeric Powder collectively account for the majority of sales and are classified as Category A.
- Cumin Seeds, Chilli Powder, and Yellow Mustard Seeds fall into Category B, contributing moderately to sales and requiring balanced inventory management.
- Brown Mustard Seeds and Fennel Seeds are Category C, with minimal contribution, and therefore need only basic stocking and limited managerial focus.

Average Fulfilment Rate of all Spices across 10 Districts:

	Average Fulfilment Rate							
Row Labels	Poppy	Cumin	Brown	Yellow	Fennel	Panch	Turme	Chilli
BANKURA	99.4	97.74	93.26	99.58	98.7	98.7	98.4	100
BIRBHUM	98.6	98.41	96.47	98.61	97.5	99.1	98.7	99
HOOGHLY	97.7	98.91	97.2	99.71	97.8	99.1	99.7	100
KOLKATA	99.9	98.55	92.1	98.43	98.1	98.6	99.6	99
MURSHIDABAD	98.8	97.78	95.04	98.29	98.8	98.1	99.5	99
NADIA	99.4	98.37	95.71	98.48	98.5	98.2	99.3	98
NORTH 24 PGS.	99.1	98.78	97.39	99.04	98.2	99.5	99.4	100
PASCHIM BURDWAN	98.3	98.52	96.05	98.77	97.5	98.5	99.5	100
PASCHIM MIDNAPORE	98	97.87	95.89	97.93	98.4	99.3	82.9	83
PURBA BURDWAN	99.1	98.56	95.96	98.63	97.8	99	99.5	100

- The average fulfilment percentage across all districts and pack sizes is above 90% with exceptions: Paschim Midnapore – Turmeric (82.94%), Chilli (82.64%).

Number of Products (inclusive of all pack sizes) sold across 10 Districts:

Total Sales Qty(from April'24-March'25)	
NORTH 24 PGS.	182456
HOOGHLY	127422
PURBA BURDWAN	49167
PASCHIM BURDWAN	26814
PASCHIM MIDNAPORE	21851
NADIA	20935
BIRBHUM	16084
BANKURA	11004
MURSHIDABAD	10088
KOLKATA	8729

- North 24 Pgs, a major distribution hub, contributes to 46% of total predicted sales. Hooghly also performs strongly, about 32% of total sales volume.

Analogous Product Analysis

Market prices of comparable brands (JK, Sunrise, Ganesh) were studied for 50 g packs

Corindo Spices Ideal Price Suggestion (50g pack)

Product	Min Price	Max Price	Average Price	Ideal_Competitive_Price
Shahi Garam Masala	₹ 90.00	₹ 130.00	₹ 116.67	₹ 105.00
Kashmiri Lal Mirch	₹ 60.00	₹ 70.00	₹ 66.67	₹ 60.00
Coriander Powder	₹ 20.00	₹ 20.00	₹ 20.00	₹ 18.00

- Shahi Garam Masala has the highest price variation while Coriander Powder is highly commoditized (all competitors price the same).
- Competitive pricing is set slightly below the market average to attract early buyers.

The average monthly orders for 50 g packs of Chilli Powder and Cumin Powder were found to be 404 units and 520 units, respectively. These existing products were used as benchmarks to estimate the likely demand for Corindo Spices' upcoming products.

Assuming that ABC Spices will initially capture around 20% of the existing market for similar spices:

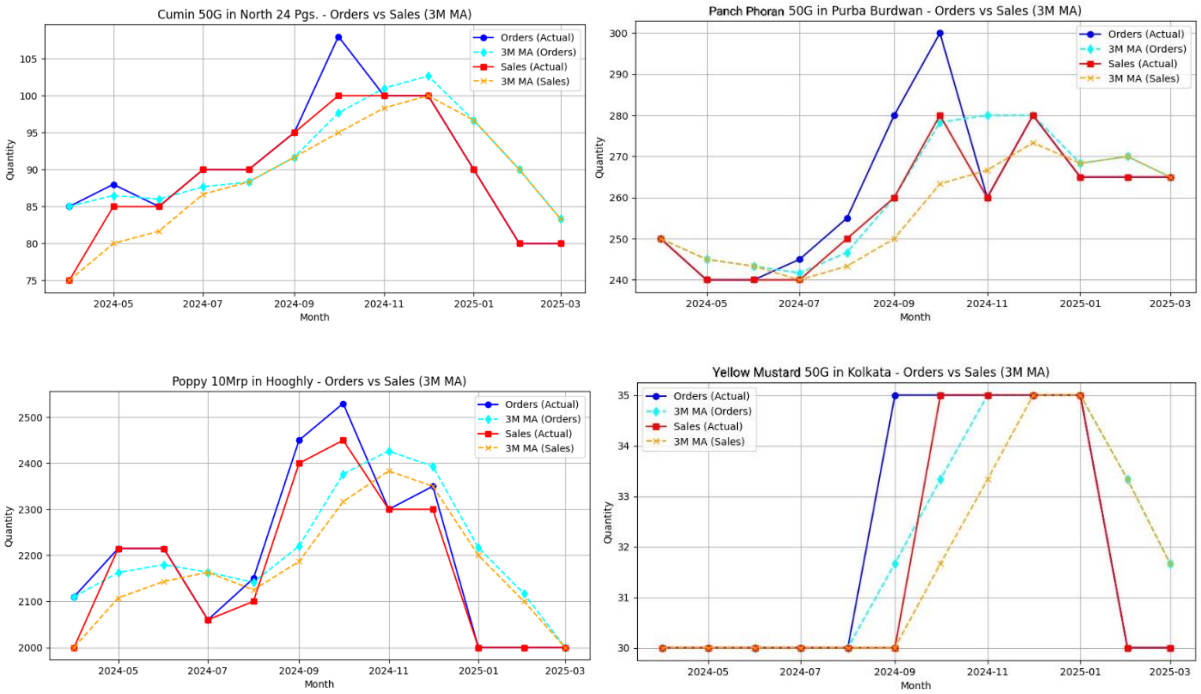
$$\textit{Expected Orders} = \textit{Average Monthly Sales of Analogous Product} \times 0.2$$

$$\textit{Initial Stock} = \textit{Expected Orders} \times (1 + \textit{Safety Stock \%})$$

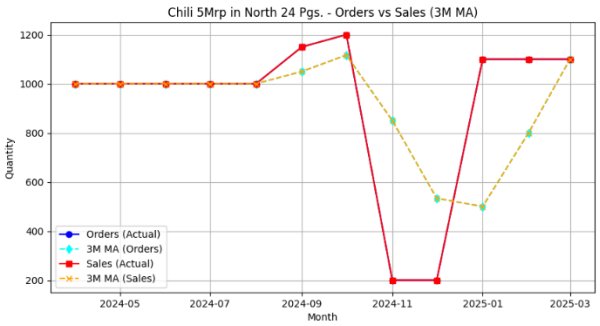
Product	Expected Orders	Safety Stock	Initial Stock
Shahi Garam Masala	81	20%	97
Kashmiri Lal Mirch	81	20%	97
Coriander Powder	104	20%	125

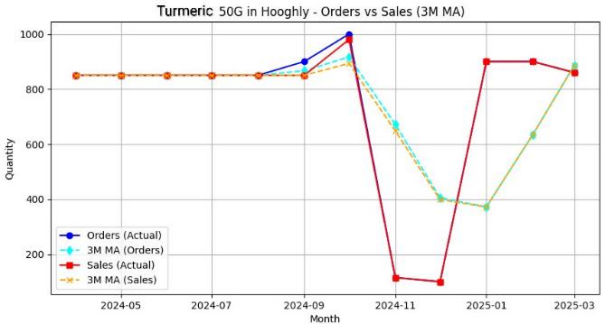
- Corindo Spices should prioritize launching the three new products in top-performing districts, such as North 24 Parganas and Hooghly, which demonstrate high order volumes and strong average fulfilment rates.

Moving Average Analysis



- Most spices show a rise in sales during festive seasons, reflecting increased household and wholesale demand.
- Additionally, Poppy and Fennel powders peak during the summer months, likely due to seasonal recipes and consumption patterns.





- Interestingly, Chilli Powder and Turmeric Powder with a spike during the onset of festive season, show unusual and sharp dip in Nov 2024, followed by a quick recovery.

In focus: unusual dip in Turmeric and Chilli sales in November-December 2024

- *Meeting with the owner*- Upon discussion with the business owner, it was revealed that in October, quality issues were reported for both Turmeric and Chilli Powder, which subsequently led to decreased orders across several districts. The products were noted to become moist and lose their freshness (likely due to poor packaging).
  The impact was particularly sharp in Hooghly, North 24 Parganas, Paschim Midnapore, and Purba Burdwan for Turmeric, and in Hooghly, North 24 Parganas, and Purba Burdwan for Chilli.
  The company then approached a more secured and suitable packing for these two spices and reassured the wholesalers and retailers of the standard quality of spices.